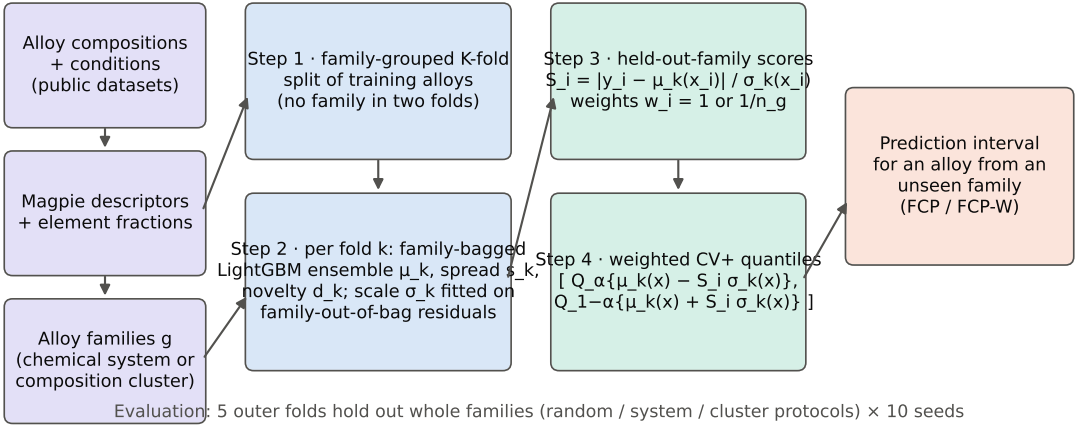
Alloy compositions + conditions (public datasets)
Magpie descriptors + element fractions
Alloy families g (chemical system or composition cluster)
Step 1 · family-grouped K-fold split of training alloys (no family in two folds)
Step 2 · per fold k: family-bagged LightGBM ensemble μ_k, spread s_k, novelty d_k; scale σ_k fitted on family-out-of-bag residuals
Step 3 · held-out-family scores S_i = |y_i − μ_k(x_i)| / σ_k(x_i) weights w_i = 1 or 1/n_g
Step 4 · weighted CV+ quantiles [Q_α{μ_k(x) − S_i σ_k(x)}, Q_1−α{μ_k(x) + S_i σ_k(x)}]
Prediction interval for an alloy from an unseen family (FCP / FCP-W)
Evaluation: 5 outer folds hold out whole families (random / system / cluster protocols) × 10 seeds